

Evaluating Plausibility in VR Beach Volleyball with varying levels of Game and Crowd Population

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Supervisor: Dr. Gareth Young

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...ABSTRACT...

Acknowledgments

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Chapter 1

Introduction

Athletes in professional, highly competitive sports have every interest in resorting to the newest advances in technology in order to enhance their competitive performance, and training regime. With recent jumps in technology, Virtual Reality (VR) is increasingly establishing itself as a valuable tool to professional athletes in training scenarios, thanks to its ability to display a wider variety of settings and activity, at a higher level of control than training grounds ever could grant on a regular basis.

Earliest adoptions of VR, or immersive technologies in professional sport include 3D recreations of matches for more immersive match reviews from either first-person, or 3-dimensional non-fixed viewpoints in the early 2010s Farley et al. (2020).

The 2022 inews article "Half of Premier League clubs use virtual reality to heal injuries and recreate matchday pressure in training" Cunningham (2022) highlights the increasing popularity of VR across the elite-level football, or soccer, scene, with half of the twenty English Premier League teams of the time applying VR scenarios for injury rehabilitation, recreation of matchday atmosphere and pressures, and other training and academy uses, which were left more vague.

The advantages of this technology are manifold: besides the proven ability to positively affect performance and motivation through various different factors interacting together, e.g. immersion (Ijsselstein et al. (2004)), competitiveness (Nunes et al. (2014)), digital cooperation (Irwin et al. (2012)), agency in selecting and adjusting the environment (Neumann et al. (2018)), or adaptive difficulty levels (Gray (2017), Richlan et al. (2023)), it is a technology that allows for safer training conditions, exemplified with Ritter et al. (2023) showcasing that gymnasts practising on a beam at floor level - determined to be a safer training environment for the study than the physical facility featuring a beam raised above the floor - whilst wearing a Head-Mounted Display (HMD) experienced similar psychological conditions to the control group practising in the physical environment.

VR is today also applicable at a small scale, e.g. in local gyms, or in dedicated spaces at home or in training centres, and affordably through commercial-grade HMDs Farley et al. (2020).

With the increased interest, there is a rise in scrutiny of which ways the incorporation of VR can benefit athletes, with potential for use in sharpening motor skills such as movement and coordination, cognitive skills like perception on the pitch or positioning, mental skills to better resist match or score pressure, or tactical and strategic proficiency in helping to understand and apply directives from the coach Richlan et al. (2023). Newer research, such as Ritter et al. (2023) and Gerwann et al. (2025), also promotes the introduction of Augmented Virtuality (AV), which introduces interaction with physical objects to provide or simulate haptic feedback whilst performing tasks in VR.

However, despite the widening adoption of the technology and increased research, the factors affecting performance in, and skill acquisition through VR are still a very young field of research that is yet to accurately study the aspects in a more granular and standardised fashion, as various reviews of existing studies into the topic of performance in VR sports applications, and their effect on competitive output, highlight Neumann et al. (2018) Richlan et al. (2023) Gerwann et al. (2025).

Despite a notable extension of the range of sports and sport tasks studied between the reviews of Neumann et al. (2018) and Richlan et al. (2023), the former of which advocated for exactly this extension beyond mostly aerobic workout activities, a sport that seems conspicuously absent from studies so far is Volleyball, alongside any of its variations, and probably for good reasons, as it requires dealing with many inherent limitations of VR application to sport, outlined in Richlan et al. (2023).

That being said, there is ample evidence that the acquisition and refinement of complex skills in sport does not require a perfect recreation of the environment, and can be compartmentalised to fit specific purposes at a time, as evidenced by Farley et al. (2020)'s proposal of a system making use of a skateboard with VR environments, to practise surfing in dry conditions for example. Therefore, volleyball seems a suitable candidate for further investigation.

This dissertation therefore aims to stimulate further research into applying VR to volleyball applications by investigating whether, and in what way, simulated crowds meaningfully affect, both positively or negatively, users' sense of immersion or, more specifically,

the illusion of plausibility that can make or break this feeling of immersion - itself a primordial factor in promoting the plethora of positive effects made possible by interactive VR.

Background knowledge required to grasp the various aspects of this study will be outlined before a selection and comparison of existing literature on, or related to, the subject in Chapter 2.

The subject of the study, and proposed solution theretowards will be detailed in Chapter 3, while the technical contributions made throughout this project are showcased in Chapter 4.

The proceedings of the study and its results will be looked into in Chapter 5, before closing this dissertation with a critical look at the study and its results, and suggestions for directions research building upon this paper might be worth taking in Chapter 6.

Chapter 2

State of the Art

At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

From the perspective of this document forming part of your degree, this chapter should demonstrate to the reader your knowledge of the area of your dissertation project. It should present your knowledge in a coherent and detailed form. The reader should understand that you have in-depth knowledge of the area of the dissertation without being overloaded with information.

2.1 Background

A section on the background of the dissertation should provide the reader with an introduction to existing technologies and concepts that form the basis of the work presented in the dissertation.

2.2 Closely-Related Work

Work in research areas tends to address a number of specific aspects. Ideally, the discussion of published research should focus on the aspects that have been addressed by various publications - and not a discussion of the individual publications.

For example, if the topic would be a discussion of work on programming languages, the subsections of the related work could be discussions of object orientation and its realisation in various languages or the use of lambda functions by these languages.

2.2.1 Aspect #1

2.2.2 Aspect #2

2.3 Summary

Summarize the chapter and present a comparison of the projects that you reviewed.

	Aspect #1	Aspect #2
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 2.1: Caption that explains the table to the reader

Chapter 3

Design

ultimate goal of this project is to propose avenues of further research specific to promoting immersion through plausible crowds. (more details needed)
maybe reexplain choice of (beach) volleyball x crowd population

structure of chapter:

- Problem Formulation
 - Identified Challenges (gaps in literature),
 - Proposed Work (RQ & rough description of solution)
- Overview of the design ($\sim$ Methodology)
- Summary of the section

3.1 Problem Formulation

This section should provide the reader with an overall description of the problem that will be addressed in the dissertation. In contrast to a generic discussion of the dissertation topic in the Introduction chapter, this section should provide a detailed discussion of the problem that has been identified based on the existing work that has been discussed in the preceeding chapter.

In some dissertations, it may make sense to convert this section into a short chapter of its own which follows the discussion of the existing work and preceeds the discussion of the work of the dissertation.

3.1.1 Identified Challenges

This section should present a short description of the gaps in the existing work and the relationship of these gaps to the work described in this dissertation.

In all reviews, I have never seen Volleyball mentioned as an example, probably because of the huge challenges to reliably portraying and executing VB in iVR.

3.1.2 Proposed Work

This section should provide a thorough description of the problem and an overview of the work proposed to address the problem.

3.2 Overview of the Design

A description of the approach that addresses the problem identified above.

Hardware: standalone Meta Quest 3 with two controllers, one per hand. -¿ standalone to accommodate for free arm movement, and 360° physical turning.

Room with small, open windows, allowing for ~ 4x4m unobstructed space for participants to play in. Researcher present in the research space to redirect towards the centre, should they stray a bit too close to an obstacle, and make private observations based on exhibited behaviour and speech. All experiments done in the same room at different times of day throughout a 2 week period. 20 mins minimum play time including freeplay.

Avatar & body: generic VRMP avatar, never rendered or mirrored to participants. Shown hands a generic model that should be ambiguous w.r.t. gender, with blue chosen as skin tone for neutrality in that regard. Hands were not animated whatsoever, always presenting an open palm. No legs for practical reasons.

Participants: n participants recruited through email and personal connections -¿ bias in population towards university students, mostly male. Ethics tangent: list up flow of participation + ethics approval by the committee for SCSS @ TCD + unpaid.

Note on questionnaire: 1-10 ratings are called a Likert scale.

3.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 4

Implementation

Focus of this chapter on detailing the game built for the study, as the object used to conduct the study. Presented in [x] order.

structure of chapter:

- Overview (Detail how there wasn't a satisfactory, commercially available VR VB app for the study, so had to make my own. List requirements.)
- Technical components
 - VRMP template + assets
 - Crowds - visuals, audio, crowd levels
 - match management + implemented rules + ball lifetime
 - short section on ball physics (citing Unity documentation on basic physics)
 - Hitting physics - iterations from built-in physics, velocity approximation, then sensor collection
 - hitting mechanics - aim estimation, hit strength modifiers, ways of application.
 - Incoming ball mechanic.
- Summary of the section

4.1 Overview of the Solution

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 4.1: Lengthy caption explaining the code to the reader

The code in listing 4.1 is a demonstration how to include a file with code into the template.

4.2 Component One

The code in listing 4.2 is a demonstration how to include code in the template.

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 4.2: Second Lengthy caption

Virtual crowd from plugin (add ref here), modified for: a) randomised seating (! not at runtime, every participant had the exact same crowd), and b) dynamic density adjustment. Audience was composed of a random choice between n different spectator prefabs, each assigned a distinct animation rooted in a different sentiment, and all wearing a different colour shirt. Spectators were diverse* in both gender and ethnicity, and were recognisably Human but did not have any faces, thus couldn't have any facial expressions, any reaction was based solely on gestures.

get framerate from app whenever trials are done.

4.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 5

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

5.1 Experiments

In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 5.1.

Column 1	Column 2	Column 3
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 5.1: Caption that explains the table to the reader

5.2 Results

Figures that present results such as figure 5.1 need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.

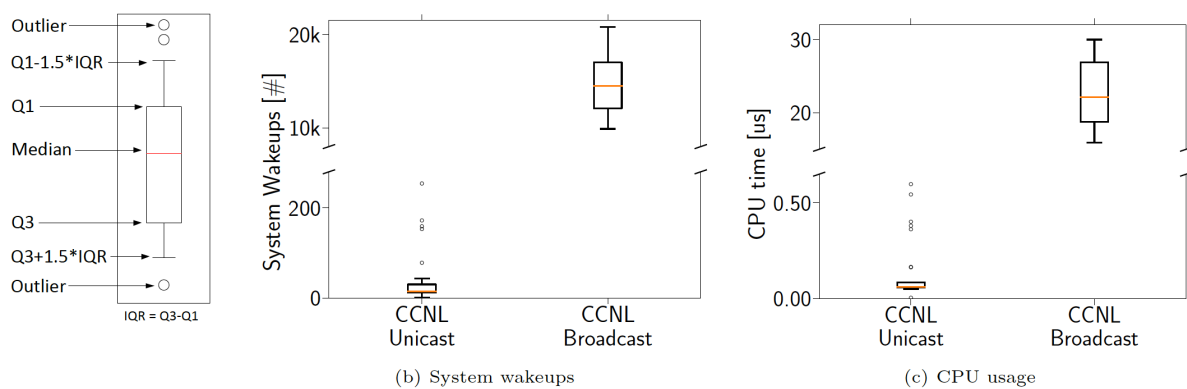


Figure 5.1: Long caption that describes the figure to the reader

5.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 6

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter 5.

6.1 Limitations

- VB in VR exposes MANY limitations of iVR in general:
 - complete absence of haptic feedback
 - dangerosity from lack of vision of feet/ground
 - need for micro-detail recognition (ex. for blockers, any indication, e.g. eye or head movement, hand orientation and signals, from the setter where the ball will go counts)
 - very complex ball handling compared to other sports (uniquely so)
 - complexity of implementation to a realistic level (many nitpicky rules e.g. double touching, holding the ball for too long during a set, setting foot on the other side of the net by accident etc.)
- simplicity of implementation + faulty aim estimation & hit classification

6.2 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

Game improvements:

- use of controller-free controls (e.g. hand-tracking, maybe full-body or foot-tracking for more holistic games) - e.g. for specialised systems through a real-time MoCap setup - could be paired with haptic gloves to simulate ball physics on hands!
- Better hit classification (esp. from one hand), refining aim recognition for specific mechanics (e.g. dig particularly)
- including actual opponents rather than just reacting to balls flying in

Study direct improvements:

- use as a learning experience, e.g. using this to teach complex IRL mechanics (e.g. setting) and measuring how well that translates to improvement on the pitch.
- more in-depth probing for results (e.g. sentiment analysis on interviews or similar, more open responses, as lauded by Slater in Slater et al. (2023) to pick up on a much larger depth of information than a 10 point scale ever could)

Tangential directions for further research:

- More alive/reactive crowds (with more accurate levels of engagement, maybe recorded from events of the actual sport; also more individual shouts, more reactivity to on-pitch happenings (oohs and ahhs with bigger events e.g. blocks), or wider range of emotion - cheering, booing, jeering, chants, boredom/relative quiet etc.), e.g. using researched crowd behaviour models like the one used for Glémarec et al. (2021).
- more detailed investigation into human factors (what kind of trait reacts the most to the taunting spectator, what kind of trait is associated with complete indifference from the crowd etc.)

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Appendix

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